



MANIPAL
ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

**Master of Engineering - M.E in Computer Science and
Engineering**

Syllabus

July 2026 Onwards

**MANIPAL SCHOOL OF INFORMATION SCIENCES,
MANIPAL ACADEMY OF HIGHER EDUCATION,
MANIPAL - 576104, KARNATAKA, INDIA.**



Program Educational Objectives / Outcomes (PEOs)

PEO1: Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.

PEO2: Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.

PEO3: Promote scientific and societal advancement through research and entrepreneurship.

Program Objectives / Outcomes (POs)

PO1: An ability to independently carry out research / investigation and development work to solve practical problems.

PO2: An ability to write and present a substantial technical report/document.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.

PO4: An ability to develop software based on theoretical principles and practical knowledge.

PO5: An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



Program Structure

Semester I									
Course Code	Course Name	No. of Hours/week				Duration of Exam in Hrs	Maximum Marks		
		Lecture	Tutorial	Practical	Credit		Internal 50	External 50	Total 100
AME 5002	Applied Machine Learning	3	-	-	3	3	50	50	100
AME 5101	Applied Linear Algebra	3	-	-	3	3	50	50	100
CCE 5101	Cloud Application with JAVA and Database	3	-	-	3	3	50	50	100
ESE 5102	Data Structures and Algorithms	3	-	-	3	3	50	50	100
ESE 5104	Real Time Operating Systems	3	-	-	3	3	50	50	100
AME 5052	Applied Machine Learning Lab	-	-	3	1	3	50	50	100
AME 5151	Applied Linear Algebra Lab	-	-	3	1	3	50	50	100
CCE 5151	Cloud Application with JAVA and Database Lab	-	-	3	1	3	50	50	100
ESE 5152	Data Structures and Algorithms Lab	-	-	3	1	3	50	50	100
ESE 5154	Real Time Operating Systems Lab	-	-	3	1	3	50	50	100
MPT 5100	Mini Project - I	-	-	-	4	-	100	-	100
PSD 5100	Professional Skill Development - I	-	-	-	1	-	100	-	100
Total		15	-	15	25				



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Semester II									
Course Code	Course Name	No. of Hrs. / week				Duration of Exam in Hrs	Maximum Marks		
		Lecture	Tutorial	Practical	Credit		Internal 50	External 50	Total 100
AME 5202	Deep Learning	3	-	-	3	3	50	50	100
BDE 5202	Modern Databases for Big Data	3	-	-	3	3	50	50	100
ESE 5201	Device Drivers	3	-	-	3	3	50	50	100
	Elective - I	3	-	-	3	3	50	50	100
	Elective - II	3	-	-	3	3	50	50	100
AME 5252	Deep Learning Lab	-	-	3	1	3	50	50	100
BDE 5252	Modern Databases for Big Data Lab	-	-	3	1	3	50	50	100
ESE 5251	Device Drivers Lab	-	-	3	1	3	50	50	100
	Elective - I Lab	-	-	3	1	3	50	50	100
	Elective - II Lab	-	-	3	1	3	50	50	100
MPT 5200	Mini Project - II	-	-	-	4	-	100	-	100
PSD 5200	Professional Skill Development - II	-	-	-	1	-	100	-	100
TOTAL		15	-	15	25				

Semesters III & IV									
CSE 6098	Project Work	-	-	-	30				
Total Number of Credits to Award Degree							80		



List of Electives (Theory)

Elective - I		Elective - II	
Course Code	Course Name	Course Code	Course Name
AME 5204	Applied Computer Vision	AME 5003	Principles of Natural Language Processing
BDE 5203	Data Engineering	AME 5205	Advanced Generative AI and Agentic Systems
CCE 5001	DevOps for Cloud	CCE 5241	Cloud Operations
CCE 5203	Cloud Security Essentials	CYE 5203	Machine Learning for Cybersecurity
CYE 5202	Web Security and Secure Coding	ESE 5244	IT Project Management
ESE 5234	Entrepreneurship		

List of Electives (Lab)

Elective - I		Elective - II	
Course Code	Course Name	Course Code	Course Name
AME 5254	Applied Computer Vision Lab	AME 5053	Principles of Natural Language Processing Lab
BDE 5253	Data Engineering Lab	AME 5255	Advanced Generative AI and Agentic Systems Lab
CCE 5051	DevOps for Cloud Lab	CCE 5291	Cloud Operations Lab
CCE 5253	Cloud Security Essentials Lab	CYE 5253	Machine Learning for Cybersecurity Lab
CYE 5252	Web Security and Secure Coding Lab	ESE 5294	IT Project Management Lab
ESE 5284	Entrepreneurship Lab		



Semester I

AME 5002: Applied Machine Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply different types of supervised machine learning algorithms to practical problems.

CO2: Apply different types of unsupervised machine learning algorithms to practical problems.

CO3: Analyse different types of machine learning paradigms.

Unit I

12 hrs

Introduction to Machine Learning: Decision Trees, Linear Models: Overview of Supervised (regression and classification), unsupervised (clustering and dimensionality reduction), semi-supervised, and reinforcement learning with practical examples - Machine learning nomenclature: raw data, types of features and outputs, feature vector, Handling missing data, encoding categorical variables, feature scaling and normalization

Decision tree model of learning: Classification and regression using decision trees, splitting criteria, entropy, information gain, Gini impurity - Overfitting & Pruning in decision trees.

Unit II

12 hrs

Linear regression: Model, metrics (MSE, RMSE, R-square), estimation, and interpretation of coefficients, Introduction to bias/variance trade-off, regularization methods (Ridge and Lasso).

Logistic regression for binary classification: Sigmoid function, decision boundary, model interpretation.

Model evaluation and validation: Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.

Nearest neighbours' algorithms - Probabilistic modelling: Nearest neighbours' algorithms - Probabilistic modelling of data using parameters, Introduction to maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification.

Unit III

12 hrs

Ensemble Methods, Clustering and Dimension Reduction.

Ensemble learning methods: Bagging and Random Forest, Boosting techniques. Introduction to hyperparameter tuning and model improvement.

K-means clustering - computing distances and similarities, limitations, Hierarchical clustering-dendrogram construction, types of linkage, Dimension reduction using principal component analysis (PCA)-concept, mathematics and applications.

Gaussian Mixture Models (GMM): probabilistic clustering, mixture of Gaussian distributions, expectation-maximization (EM) algorithm for parameter estimation.



References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at <https://www.statlearning.com/>
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at <https://www.manning.com/books/grokking-machine-learning>
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III - Online resource available at <http://ciml.info/>
5. Module: Introduction to Machine Learning (<https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html>).
6. Learning path: Understand data science for machine learning (<https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/>).

AME 5101: Applied Linear Algebra

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply vector operations for similarity learning.

CO2: Analyse matrix representations in AI & ML models.

CO3: Apply matrix calculus for optimization in learning systems.

Unit I

10 hrs

Vector Operations & Geometry in AI Systems

Algebraic & geometric view of vectors; Dot product & similarity; Norms & regularization; Cosine similarity; Distance metrics; Standardization vs normalization; High-dimensional embeddings; Vector similarity in NLP; Feature representations; Vectorized computation.

Unit II

14 hrs

Matrix Representations & Transformations: Matrix operations; Matrix-vector product; Matrix-matrix product; Forward propagation in deep learning; Batch computation; Elementary row operations and RREF; Solving systems of linear equations; Linear dependence and independence of vectors; Rank & subspaces; Matrix factorization: attention mechanism and low-rank representations; projections and dimensionality reduction using PCA.

Unit III

12 hrs

Matrix Calculus for Learning Systems and visual: introduction to sensitivity & gradient in one dimension; Chain rule; Gradient in multiple dimensions; Matrix derivatives; Optimization using



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

gradient descent; Batch vs stochastic learning; Backpropagation for binary and multiclass classification and regression.

References

1. Introduction to Applied Linear Algebra, Vectors, Matrices, and Least Squares, Stephen Boyd & Lieven Vandenberghe, Cambridge University Press, 1st Edition, 2018. Available online at <http://vmls-book.stanford.edu/vmls.pdf>.
2. Linear Algebra and Learning from Data, Gilbert Strang, Cambridge Uni. Press, 1st Edition, 2019.
3. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
4. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
5. Neural Networks and Deep Learning, Michael A. Nielsen, Determination Press, 2015. Available online at <http://neuralnetworksanddeeplearning.com/>
6. Matrix Methods: Applied Linear Algebra, Richard Bronson and Gabriel B. Costa, Academic Press, 3rd Edition, 2008.
7. Introduction to computational thinking (<https://computationalthinking.mit.edu/Fall24/>).

CCE 5101: Cloud Application with JAVA and Database

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply Java programming techniques and data interface concepts for application development.

CO2: Analyse the techniques to be used for high availability and fault tolerant cloud applications.

CO3: Understand the Cloud application architecture and design principles.

Unit I

16 hrs

Introduction to OOPS: OOPS - Procedural vs Object Oriented languages - Abstraction - Encapsulation - Polymorphism - Inheritance.

Introduction to JAVA: JAVA Features - Present JAVA language and- JVM - JVM Architecture - JAVA Datatypes, Variables, Arrays- JAVA Basic Constructs.

Class Concepts: Objects - Methods - Revisiting Inheritance - Multilevel - Method Overriding - Abstract Class - Interface -Package-IO.

Unit II

14 hrs

Introduction to DBMS: File Systems Organization - Sequential, Pointer, Indexed, Direct - Purpose of Database System- Database System Terminologies-Database Characteristics Data models - Types of data models - Components of DBMS- Relational Algebra. | Logical database design: Relational DBMS -



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Codd's Rule - Entity Relationship model - Extended ER Normalization - Functional Dependencies, Anomaly- 1NF to 5NF- Domain Key Normal Form - Denormalization.

Java Application using Database (JDBC) Loosely coupled application development - Standalone application using JDBC - illustrate converting the same application to a web application reusing the application components.

Unit III

6 hrs

Introduction to Cloud database: Evolution of Database - Introduction to Cloud Database - Data Lake - Database as a Service - Type of Data w.r.t Big Data - Type of Cloud Databases - Introduction to NOSQL - NOSQL vs SQL - Structured, Semi structured, Unstructured data.

Introduction to cloud application development: Design Considerations for cloud Applications: Scalability - Reliability & Availability - reference Architecture for Cloud Applications - cloud Application Design Methodologies: Service Oriented Architecture - Cloud Component Model - Services of cloud Applications - Model View Controller - Restful Web Services - Data Storage Approaches: SQL - NOSQL Approaches.

References

1. Archdeep Bahaga, Vijay Madiseti, "Cloud Computing-A hands on Approach", Universities Press, 2014.
2. Patrick Naughton and Herbert Schildt, "JAVA 2 - The Complete Reference 7th & 8th Edition", Tata McGraw Hill, 2012.
3. Paul Deitel, Harvey Deitel, "How to program with JAVA", 9th Edition, Deitel Publications, 2012.
4. Abraham Silberschatz, Henry F. Korth and S. Sudharshan, Database System Concepts, Sixth Edition, Tata Mc Graw Hill, 2011.
5. C.J.Date, A.Kannan and S.Swamynathan, An Introduction to Database Systems, Eighth, Edition, Pearson Education, 2006.
6. Atul Kahate, Introduction to Database Management Systems, Pearson Education, New Delhi, 2006.
7. Alexis Leon and Mathews Leon, Database Management Systems, Vikas Publishing House Private Limited, New Delhi, 2003.
8. Raghu Ramakrishnan, Database Management Systems, Fourth Edition, Tata Mc Graw Hill, 2010.
9. Articles, White papers and Transaction Papers.
9. <https://www.coursera.org/learn/database-design-postgresql>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

ESE 5102: Data Structures and Algorithms

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Analyse the implementations of elementary data structures.

CO2: Design and implementation of data structures like trees and graphs.

CO3: Analyse the concepts of divide and conquer, dynamic programming, greedy, and back tracking techniques to efficiently solve problems.

Unit I

12 hrs

Introduction: Algorithm Specification, Performance Analysis.

Algorithm Analysis Techniques: Analysis of Recursive Programs, Solving Recurrence Equations, General Solutions for a Large Class of Recurrences, Randomized Algorithms.

Elementary data structures: Implementation of Lists, Stacks, Queues.

Unit II

15 hrs

Sorting & Searching Techniques: Quick sort, Heap sort, Merge sort, Binary search, linear search, Fibonacci search.

Trees: Basic Terminology, Implementation of Trees, Binary Trees, Binary Search Trees, Types of balanced trees - construction - and its operation.

Graphs: Basic definitions, Representation of Graphs, Minimum Cost Spanning Tree, Single Source Shortest Paths, All-Pairs Shortest Path.

(MOOC: <https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-algorithms#enroll>).

Unit III

9 hrs

Operations on Sets: Introduction to Sets, A Linked- List implementation of Set, The Dictionary, The Hash Table Data Structure.

Algorithm Design Techniques: Divide-and-Conquer Algorithms, Dynamic Programming, Greedy Algorithms, Backtracking.

References

1. "Introduction to Algorithms" Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.
2. "Data Structures & Algorithms" Aho, Hopcroft and Ullmann.
3. "Data structures and algorithm analysis in C" Mark Allen Weiss.
4. "Computer Algorithms" : Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran.
5. MOOC: <https://in.coursera.org/learn/algorithms-on-graphs?specialization=data-structures-algorithms#enroll>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

ESE 5104: Real Time Operating Systems

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Evaluate the features of OS, RTOS, processes and process scheduling algorithms.

CO2: Analyse various techniques for addressing synchronization, deadlock and memory management.

CO3: Evaluate schedulability in real time systems.

Unit I

15 hrs

Introduction to OS and RTOS: Essential features of an OS, Single Processor Systems and Multiprocessor Systems, Essential Features of Batch Processing, Time sharing, Multiprogramming, Interactive systems, User mode and Kernel Mode operations, Distinction between function call and system call, Real time operating system and real time embedded systems.

Process Management: A process in memory, process state, PCB, Process scheduling, scheduling Queues, Types of schedulers, Process system calls - IPC using Shared Memory, IPC using Sockets.

Process Scheduling: Introduction, scheduling criteria, scheduling Algorithms - FCFS, SJF, PS, RR, Multilevel Queues, Multilevel feedback Queue Scheduling, Scheduling evaluations.

Unit II

16 hrs

Multithreaded Programming: Introduction, benefits, multithreading models, Pthreads, Win32 threads, Threading Issues, Thread pools Linux threads.

Synchronization: Introduction, Critical Section Problem, Petersons Solutions, synchronization hardware, Semaphores, usage, implementations; Deadlocks and starvation, Classical problem of synchronization - Bounded Buffer problem, Reader's Writer's problem, Dining Philosophers problem, sleeping barbers problem; Monitors.

Deadlocks: Introduction, deadlock, characterization, methods for handling deadlocks, deadlock prevention, deadlock avoidance, recovery from deadlock.

Memory Management: Memory Management Strategies, Virtual Memory Management.

Unit III

5 hrs

Real Time Systems: Overview of Real Time Systems, Real Time clocks and Real Time Scheduling Algorithms - RMS, DMS, EDF, priority inheritance.

(MOOC:<https://in.coursera.org/learn/real-time-embedded-systems-concepts-practices#syllabus>).

References

1. "Operating System principles", Seventh Edition, Abraham Silberschatz, Peter Galvvin, Grag Gagne. John Wiley Publications.



2. “Real - Time Systems and Programming Languages”, Allan Burns, Andy Wellings.
3. “Operating Systems Concepts and Design”, Milan Milenkovic.
4. “Design of Unix Operating System”, Maurice Bach (IPC)
5. “The C Programming Language”, Kernighan & Ritchie.
6. MOOC: <https://in.coursera.org/learn/real-time-embedded-systems-concepts-practices#syllabus>.

AME 5052: Applied Machine Learning Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply different types of supervised machine learning algorithms to practical problems.

CO2: Apply different types of unsupervised machine learning algorithms to practical problems.

CO3: Analyse different types of machine learning paradigms.

Unit I

12 hrs

Introduction to Machine Learning, Decision Trees, Linear Models: Basics of Python for data handling: Introduction to Python libraries for machine learning (NumPy, Pandas, Matplotlib, Scikit-learn). Understanding datasets and data preprocessing techniques such as handling missing values, feature scaling, and encoding categorical variables.

Decision tree model of learning: Implementation of classification and regression using decision trees. Application of splitting criteria such as entropy, information gain, and Gini impurity. Analysis of overfitting and implementation of pruning techniques.

Unit II

12 hrs

Linear regression: Implement linear regression model, apply different metrics (MSE, RMSE, R-square).

Logistic regression for binary classification: Implement logistic regression model, apply sigmoid function, decision boundary. Apply regularization methods (Ridge and Lasso).

Nearest neighbours' algorithms: Apply K-NN algorithms for the practical problems.

Probabilistic modelling: Implement maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification.

Model interpretation, evaluation and validation: Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.

Apply bias/variance trade-off.

Visualize, compare, and contrast different models for practical problems.



Unit III

12 hrs

Clustering techniques: Computing distances and similarities, implementation and analysis of K-means clustering and Hierarchical clustering, dendrogram construction and linkage methods.

Ensemble learning methods: Through coding, understand how ensemble methods in machine learning work, implementation and analysis of bagging techniques such as Random Forest and boosting algorithms.

Probabilistic clustering: Implementation of Gaussian Mixture Models (GMM) for clustering tasks.

Dimensionality reduction: Implementation and visualization using Principal Component Analysis (PCA).

Visualize, compare, and contrast different models for practical problems.

References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at <https://www.statlearning.com/>
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at <https://www.manning.com/books/grokking-machine-learning>
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III - Online resource available at <http://ciml.info/>
5. Module: Introduction to Machine Learning (<https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html>).
6. Learning path: Understand data science for machine learning (<https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/>).

AME 5151: Applied Linear Algebra Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply Python's legacy libraries for coding vector- and matrix-related operations.

CO2: Implement models for real-life applications using linear algebra principles and analyze the results from a practical perspective.

CO3: Implement building blocks of deep learning models efficiently using linear algebra principles.



Unit I

10 hrs

Vector Operations in Practice: Visualize vectors and vector-operations and relate them to their geometric descriptions; Implement and interpret results of distance-based clustering; Understand the importance of feature scaling; Compare words using embeddings: vector similarity computation; cosine similarity.

Unit II

14 hrs

Matrix Computation Workflows: Understand how to perform matrix operations using Python; Implement and interpret matrix-operations using block representations; implement application examples from deep learning: data and weight matrices; forward propagation using matrix-based approach; batch forward propagation; attention mechanism; Implement and interpret dimension reduction using PCA; Solve systems of linear equations arising from application problems.

Unit III

12 hrs

Optimization in Practice: Gradient descent implementation; Matrix-based optimization; Implement application examples from deep learning - backward propagation using matrix-based approach for binary and multiclass classification and regression

References

1. S. Boyd and L. Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2018. [Online]. Available: <http://vmls-book.stanford.edu/vmls.pdf>
2. G. Strang, *Linear Algebra and Learning from Data*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2019.
3. IBM, "AI Engineering Professional Certificate," Coursera. [Online]. Available: <https://www.coursera.org/professional-certificates/ai-engineer>
4. F. Chollet, *Deep Learning with Python*, 2nd ed. Shelter Island, NY, USA: Manning Publications, 2021.
5. M. A. Nielsen, *Neural Networks and Deep Learning*. Determination Press, 2015. [Online]. Available: <http://neuralnetworksanddeeplearning.com/>
6. R. Bronson and G. B. Costa, *Matrix Methods: Applied Linear Algebra*, 3rd ed. Burlington, MA, USA: Academic Press, 2008.
7. Massachusetts Institute of Technology, "Introduction to Computational Thinking," 2024. [Online]. Available: <https://computationalthinking.mit.edu/Fall24/>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

CCE 5151: Cloud Application with JAVA and Database Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Experiment with Java programming techniques and data interface concepts for application development.

CO2: Design and develop the applications with java as frontend interacting with relational database.

CO3: Examine the techniques used for fault tolerant applications.

Unit I

12 hrs

Application development using programming constructs. Set up a Database environment with proper user access levels and Demonstrate the DB operations.

Unit II

12 hrs

Functioning of JDBC concept connecting Java application to a database modelled as per application requirements.

Unit III

12 hrs

Development of Java application that demonstrates the key concepts related to cloud application - namely loosely coupled, log enabled Java application.

References

1. Archdeep Bahaga, Vijay Madiseti, "Cloud Computing-A hands on Approach", Universities Press, 2014.
2. Archdeep Bahaga, Vijay Madiseti, "Internet of Things - A hands on Approach", Universities Press, 2015.
3. Patrick Naughton and Herbert Schildt, "JAVA 2 - The Complete Reference 7th & 8th Edition, Tata McGraw Hill, 2012.
4. Paul Deitel, Harvey Deitel, "How to program with JAVA", 9th Edition, Deitel Publications, 2012.
5. Danny Coward, "Java WebSocket Programming", McGraw-Hill Education, 2014.
6. Jobinesh Purushothaman, "RESTful Java Web Services", Second Edition, Packt Publishing, 2015.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

ESE 5152: Data Structures and Algorithms Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement elementary data structures.

CO2: Design programs for sorting and searching.

CO3: Design programs for dictionary, hash tables, graphs, and trees techniques.

Unit I

12 hrs

Algorithm specification and analysis techniques: Analysis of recursive programs. Solving recurrence equations. General solution for a large class of recurrences.

Elementary data structures: Implementation of Array, lists, stacks, queues.

Unit II

12 hrs

Sorting & Searching: Bubble, selection, insertion sort, Quick sort, heap sort, merge sort. Linear search and binary search.

Unit III

12 hrs

Trees and Graph: Implement Hashing and Dictionaries, Binary Trees, Binary Search Trees. Graphs - Depth First Searching and Breadth First Searching. Minimum cost spanning tree. Single source shortest paths.

Algorithm Design Techniques: Divide-and-Conquer Algorithms, Dynamic Programming, Greedy Algorithms, Backtracking.

References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.

ESE 5154: Real Time Operating Systems Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement C programs to efficiently solve problems using multiple processes.

CO2: Execute C programs for process scheduling and deadlock avoidance algorithms.

CO3: Design multithreaded programs to solve problems considering process scheduling, deadlocks, or process synchronization on a given scenario.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Unit I

15 hrs

Conduct experiment to display the process states of various active process.

Conduct experiments by using POSIX system calls.

Unit II

6 hrs

Conduct experiment on process scheduling algorithms.

Conduct experiment to demonstrate process deadlocks.

Unit III

15 hrs

Conduct experiment to demonstrate process synchronization.

Conduct Experiment on multithread programming.

Conduct experiment to demonstrate the working of real-time operating systems using FREERTOS.

References

1. "Operating System principles", Seventh Edition, Abraham Silberschatz, Peter Galvvin, Grag Gagne. John Wiley Publications.
2. "Real - Time Systems and Programming Languages", Allan Burns, Andy Wellings.
3. "Operating Stems Concepts and Design", Milan Milenkovic.
4. "Design of Unix Operating System", Maurice Bach (IPC).
5. "The C Programming Language", Kerningham & Ritchie.

MPT 5100: Mini Project - I

(L-T-P-C: 0-0-0-4)

Total 48 hours

Course Outcome

CO1: Identify and formulate a real-world, technologically or socially relevant problem.

CO2: Design, experiment and execute the mini project effectively as a team, using appropriate tools and methodologies.

CO3: Demonstrate the ability to communicate the findings through written reports and oral presentations.

Unit I

6 hrs

Problem identification, literature survey, formation of detailed specifications.

Unit II

30 hrs

Design and implementation of the proposed system.

Unit III

12 hrs

Presentation and demonstration of the work carried out to a panel of experts.



References

1. Research Methodology, Methods and Techniques, C. R. Kothari; New Age International Publishers; 2nd Revised Edition, 2004
(<https://www.jsscacs.edu.in/sites/default/files/Files/Statistics%20and%20Research%20by%20Kothari.pdf>).
2. Project Management by Adrienne Watt - BCcampus Open Textbook project
(https://www.opentextbooks.org.hk/system/files/export/15/15694/pdf/Project_Management_15694.pdf).
3. Research articles and Online Resources.

PSD 5100: Professional Skill Development - I

(L-T-P-C: 0-0-0-1)

Total 12 hours

Course Outcome

CO1: Assemble important themes in the field of engineering domain and present a topic for effective communication in oral and written forms.

CO2: Demonstrate self-learning capabilities through certification in the domain of study.

Unit I

7 hrs

Technical Seminar and Report Writing: Presenting a technical seminar in the classroom to an audience, where the speaker is assessed on the content delivered, conceptual understanding, and presentation skills such as audibility, eye contact, and memory. The report writing component includes identifying a topic of current relevance in the field of engineering, technology, or interdisciplinary domains. Students are expected to structure the report logically, write a clear abstract, develop well-organized content, and conclude with future scope. Proper citation of sources and preparation of a bibliography are essential aspects of the report.

Unit II

5 hrs

Micro-Credential:

Impactful Writing Skills:

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_WADHWANI_03

(or)

Python Fundamentals:

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIGI_08

(and)

Linux Fundamentals

<https://www.coursera.org/learn/linux-fundamentals>

(or)



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

GitHub Foundations:

<https://learn.microsoft.com/en-us/collections/o1njfe825p602p>

(or)

Google IT Automation with Python Professional Certificate:

<https://www.coursera.org/professional-certificates/google-it-automation#courses>

(or)

Linux Fundamentals:

<https://www.coursera.org/learn/linux-fundamentals>

(or)

AWS Academy Cloud Foundations [110120]:

<https://awsacademy.instructure.com//courses/110120>

(or)

Python fundamentals (20hr) (few hr overlapped with PDV)

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIGI_08

(or)

Interpersonal Skills - advanced (11 hr)

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_WADHW_23

(or)

AWS Academy Generative AI Foundations [118816] - Educator: AWS Academy Generative AI Foundations is intended for students who seek an introduction to artificial intelligence with a particular emphasis on generative AI technologies. It provides fundamental concepts, capabilities, principles, use cases and associated AWS services and tools of generative AI in the context of the AWS cloud.

<https://awsacademy.instructure.com//courses/118816>

Reference

1. Research articles and Online Resources.

Semester II

AME 5202: Deep Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Design and analyze deep learning models for application problems.

CO2: Formulate techniques for improving the way neural networks learn.

CO3: Select appropriate deep learning models for application problems.



Unit I

12 hrs

Deep Neural Networks: One hidden layer neural network, architecture and notation - The role of activation functions and their gradients - Loss functions: binary cross entropy loss, categorical cross entropy loss, focal loss, dice loss, and weighted loss functions - Deep L-layer neural network: architecture, notation, and building blocks - The importance of deep representations - The learning slowdown problem in deep neural networks: backward propagating gradient shrinkage due to softmax and activation function gradients, vanishing and exploding gradients - Improving the way deep neural networks learn: weight initialization, L1/L2 & dropout regularizations, residual & skip connections, batch normalization, adaptive learning, learning rate scheduling, gradient clipping, early stopping, and data augmentation.

Unit II

12 hrs

Transformer Networks: Word vectors and language models: word embeddings - Transformers: single and multi-head attention mechanisms; language models; vision transformers, generative image transformers, and multimodal transformers.

Unit III

12 hrs

Advanced Topics in Deep Learning - Convolutional neural networks - Transfer Learning - Generative adversarial networks - Autoencoders - Knowledge Distillation - Quantization.

References

1. Neural Networks and Deep Learning, Michael Nielsen - Determination Press - Available online at <http://neuralnetworksanddeeplearning.com/index.html>
2. Deep Learning with Python, François Chollet, Manning Publications; 2nd Edition, 2021.
3. Deep Learning Foundations and Concepts, Christopher M. Bishop and Hugh Bishop, Springer; 2024
4. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
5. Lecture slides of Prof. Andrew Ng - Stanford University - Available online at <https://cs230.stanford.edu/syllabus/>
6. Tensor Flow for Deep Learning Paperback, Reza Zadeh & Bharath Ramsundar, O'Reilly, 2018.
7. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville - MIT Press - Available online at <http://www.deeplearningbook.org/>
8. Learning path: Microsoft Azure AI Fundamentals: Explore natural language processing (<https://learn.microsoft.com/en-us/training/paths/explore-natural-language-processing/>).



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

BDE 5202: Modern Databases for Big Data

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Analyse different types of data models and design queries to handle different data types.

CO2: Evaluate different consistency models.

CO3: Apply the knowledge of SQL and NoSQL data models.

Unit I

15 hrs

Introduction to growth of traditional and modern database systems, introduction to RDBMS.

Relational data models - Syntax and Semantics of SQL DDL and DML.

NoSQL Data Models - Document databases, Column-Family stores, vector databases.

Unit II

9 hrs

Distribution models for scalability: Horizontal partitioning. - Data sharding. -Master-slave replication. Peer-to-peer replication. - Version stamps - business and system transactions. Consistency Models: Update consistency, Read Consistency, CAP Theorem, Introduction Rest API and JSON.

Unit III

12 hrs

Case study: Key-Value databases - Document databases, MongoDB, Column-Family Stores, Graph Databases, Vector databases. Beyond NoSQL: File systems, Event sourcing, Memory Image, Version control. Choosing your database.

References

1. Database System Concepts, Avi Silberschatz, Henry F. Korth, and S. Sudarshan. McGraw Hill, 6th Edition, 2010.
2. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Pramod J. Sadalage, Martin Fowler, Addison-Wesley, 2012.
3. Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement, Eric Redmond, Jim R. Wilson, Pragmatic Bookshelf. 2012.
4. MOOCS: <https://www.coursera.org/learn/introduction-to-nosql-databases>
5. <https://www.coursera.org/specializations/cloudera-big-data-analysis-sql>

ESE 5201: Device Drivers

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Demonstrate the development of character drivers.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

CO2: Examine the importance of time and handle concurrency in device drivers.

CO3: Examine the usage of dynamic memory and communication with hardware.

Unit I

12 hrs

Introduction to Device Drivers. Building & Running Modules.

(MOOC:<https://in.coursera.org/learn/linux-kernel-programming-yocto-project#syllabus>).

Character Driver.

Debugging Techniques.

Unit II

12 hrs

Concurrency and Race Condition: Advanced Character Driver Operations, Time, Delay and Deferred Work, PCI Drivers, USB Drivers.

Unit III

12 hrs

Allocating Memory: Communicating with Hardware, Interrupt Handling.

References

1. Alessandro Rubini, "Linux Device Drivers", (Nutshell Handbook), O'Reilly Publishers, 2009.
2. John Madiou, "Linux Device Drivers Development: Develop customized drivers for embedded Linux", Packt Publishing, 2017.
3. Robert Love, "Linux Kernel Development", Addison Wesley, Third Edition, 2010.
4. Daniel P. Bovet, Marco Cesati, "Understanding the Linux Kernel", O'Reilly Media, Third Edition, 2008.
5. Wolfgang Mauerer, "Professional Linux Kernel Architecture", Wrox, 2008.
6. Sreekrishnan Venkateswaran, "Essential Linux Device Drivers", Prentice Hall, 2008.
7. W. Richard Stevens, Stephen A. Rago, "Advanced Programming in the UNIX Environment", Addison Wesley, Third Edition, 2013.
8. W. Richard Stevens, Bill Fenner, Andrew M. Rudoff, "Unix Network Programming, Vol1: Sockets", Pearson Education India, Third Edition, 2015.
9. MOOC: <https://in.coursera.org/learn/linux-kernel-programming-yocto-project#syllabus>

Elective - I (Theory)

AME 5204: Applied Computer Vision

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Understand visual feature representations in deep learning.

CO2: Analyse deep vision models for classification and detection.



CO3: Apply vision models in real-time deployment pipelines.

Unit I

10 hrs

Foundations of Visual Representation: Image representation, Basic building block of CNN, CNN Architecture, convolution-based feature extraction, deep visual embeddings, transfer learning concepts, dataset efficiency, and comparison between CNNs and vision transformers.

Unit II

14 hrs

Deep Vision Models: CNN architectures, region-based detection models, vision transformers, multimodal vision models, feature map learning, skip connections, loss functions, and fine-tuning pretrained visual models.

Unit III

12 hrs

Vision Systems in Practice: Image classification pipelines, object detection, segmentation workflows, multimodal image-text systems, latency-performance tradeoffs, edge inference, and deployment of real-time vision models.

References

1. Learning path: Microsoft Azure AI Fundamentals: Explore computer vision (<https://learn.microsoft.com/en-in/training/paths/explore-computer-vision-microsoft-azure/>).
2. Generative AI for Everyone from DeepLearning.AI (<https://www.coursera.org/learn/generative-ai-for-everyone>).
3. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
4. Lecture slides of Prof. Fei-Fei Li - Stanford University - Available online at <http://cs231n.stanford.edu/>
5. A Guide to Convolutional Neural Networks for Computer Vision, Salman Khan, Hossein Rahmani, Syed Afaq Ali Shah, and Mohammed Bennamoun, Morgan & Claypool Publishers, 2018.
6. Lecture slides of Prof. Fei-Fei Li - Stanford University - Available online at <http://cs231n.stanford.edu/>
7. Neural Networks and Deep Learning, Michael Nielsen, Determination Press - Available online at <http://neuralnetworksanddeeplearning.com/index.html>
8. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2011 - Online resource from Springer available at <http://szeliski.org/Book/>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

BDE 5203: Data Engineering

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Identify fundamental concepts of data engineering ecosystem components.

CO2: Design data engineering pipelines using data lake principles and Medallion architecture.

CO3: Develop automated data engineering workflows using orchestrators.

Unit I

10 hrs

Foundations of Data Engineering: Introduction to Data Engineering, Data Basics Refresher, Data Pipeline Concepts - Data ingestion, transformation, serving layers Data quality, validation, and monitoring basics. Modern Data Stack Overview - Data sources / Ingestion / Storage / Processing / BI/ML. Introduction to tools.

Unit II

12 hrs

Data Storage & Processing Architecture: Data Base, Data Mart, Data Warehouse, Data Lake, Delta Lake, Lake house and data ocean Concepts, Object storage, File formats. Medallion Architecture - Bronze / Silver / Gold layers, Raw / Cleaned / Aggregated, Data quality, schema evolution. Building the Pipeline Layers. Data Governance and Metadata: Data lineage, Metadata management, Data catalog concepts, Data quality frameworks (for example, Great Expectations)

Unit III

14 hrs

Orchestration with Apache Airflow: Orchestration Concepts - DAGs, tasks, operators, scheduling. DAG design patterns, Idempotency concepts, Backfills and replay strategies, Retry mechanisms, State management. Airflow architecture - Webserver, Scheduler, Executor, Metadata DB. Airflow Basics - UI tour, DAG structure, Operators, Task dependencies. Building DAG for ingestion cleaning and aggregation, Orchestrating the Medallion Pipeline - DAG for Bronze ingestion Silver transformation Gold aggregation, Scheduling, Retries, logging, failure handling.

References

1. J. Reis and M. Housley, *Fundamentals of Data Engineering: Plan and Build Robust Data Systems*. Sebastopol, CA, USA: O'Reilly Media, 2022.
2. M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*. Sebastopol, CA, USA: O'Reilly Media, 2017.
3. T. Akidau, S. Chernyak, and R. Lax, *Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing*. Sebastopol, CA, USA: O'Reilly Media, 2018.
4. M. Armbrust *et al.*, "Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics," in *Proc. Conf. Innovative Data Systems Research (CIDR)*, 2021.



5. Apache Software Foundation, "Apache Airflow Documentation," 2023. [Online]. Available: <https://airflow.apache.org>

CCE 5001: DevOps for Cloud

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Understand existing Software Methodologies and DevOps Life Cycle stages.

CO2: Analyse and implement software product development using DevOps methodologies and tools.

CO3: Apply configuration management and automation templates to provision infrastructure and configure software in DevOps lifecycle.

Unit I

16 hrs

DevOps Introduction: Understanding Development - Development SDLC: WaterFall & Agile - Understanding Operations - Dev vs Ops - DevOps to the rescue - What is DevOps - DevOps SDLC - Continuous Delivery model - DevOps tools for DevOps SDLC - DevOps Roles & Responsibilities.

Linux: Linux Introduction, Principles & Linux distro - Booting - Command line utilities & Basic commands - Linux File system - Filters & I/O Redirections - Users & Group administration - File permissions & Ownerships - Sudo - Software Management - Useful tools: ssh, telnet, scp, rsync, disk utils, backups etc - Service & Process management - Shell Scripting - Systems and HW stats - Linux Containers (lxc) - Dockers - Kubernetes and Micro services.

Networking fundamentals: Components of computer networks - Classification: LAN, WAN, Peer to Peer network, Server based - Switches - Routers - Network Architecture - Protocols - Port numbers - DNS - DHCP - IP Addresses - Ip Addresses & Subnet Masks - IP Address Ranges - Subnetting - Private Vs Public networks -High Availability - Firewalls & NACL -Web Application Architecture - Infrastructure - Network layout - Services & Components - Architecture from a DevOps perspective.

Unit II

12 hrs

Automation, Orchestration & Config Management: Version control system with Git: What is VCS & why it is needed - DevOps use cases - Setup your own repo with git - Manage your code base/source code with GIT & GITHUB.

Continuous Integration with Jenkins: Introduction to continuous integration. - Build & Release and relation with DevOps - Understanding development and developers -Why Continuous integration Jenkins introduction and setup - Jenkins projects/jobs - Jenkins plugins, Jenkins administration: Users - Nodes/slaves - Managing plugins - Managing software versions -Introduction - Phases -Java builds - Build and Release job/project setup, Nexus: Intro & Setup - Software versioning & Hosted repository - Integration with Jenkins - Continuous integration job/project setup | Complete Jenkins project: Packaging Artifacts -Static code Analysis - Tomcat setup Staging & productions - Artifacts



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

deployments to web servers from Jenkins - Build Pipeline - Jenkins not just CI tool anymore - More DevOps use cases of Jenkins.

Unit III

8 hrs

Ansible: Configuration Management & Automation - What is Ansible & its features- Ansible setup on local & cloud - Understanding Ansible architecture & Execution- Inventory, Ad hoc commands: Automating change Management with Ad Hoc commands - Playbook Introduction - Ansible configuration with ansible, cfg-Ansible documentation - Modules, modules & lots of modules -Writing playbook for webserver & DB server deployments - Tasks - Variables -Templates -Loops - Handlers - Conditions - Register - Debugging -Ansible Roles -Identify server roles - Roles structure - Creating, Managing and executing roles -Ansible Galaxy - Exploring Roles from Galaxy - Download Galaxy roles and integrate with your code - Ansible Advanced Execution - Improving execution time - Limiting and selecting tasks - Troubleshooting and Testing.

References

1. Beginning Shell Scripting (Programmer to Programmer), Eric Foster-Johnson, John C. Welch, Micah Anderson, Wrox Publications, 2005.
2. Mastering Unix Shell Scripting: Bash, Bourne, and Korn Shell Scripting for Programmers, System Administrators, and UNIX Gurus, Randal K. Michael, 2nd Edition, Wiley Publications 2nd edition, 2008.
3. UNIX & Shell Programming, Bintu Harwani, Oxford Publications, 2013.
4. Jenkins: The Definitive Guide, John Ferguson Smart, O'Reilly Media, Inc., 2011.
5. Jenkins Essentials, Mitesh Soni, Packt Publications, 2015.
6. Continuous Delivery with Docker and Jenkins, Rafal Leszko, Packt Publications, Second Edition, 2019.
7. Implementing DevOps on AWS, Veselin Kantse, Packt Publications, 2017.
8. Docker Orchestration, Randall Smith, Packt Publications, 2017.
9. Jenkins Continuous Integration Cookbook, Alan Berg, Packt Publications, 2012.
10. Practical LXC and LXD Linux Containers for Virtualization and Orchestration, Kumaran S., Senthil, A press Publications, 2017.
11. Containerization with LXC, Konstantin Ivanov, Packt Publications, Packt Publishing; 1st edition, 2017.
12. Docker: Up & Running: Shipping Reliable Containers in Production, Karl Matthias, Sean Kane, O'Reilly Media. O'Reilly Media; 1st edition, 2015
13. <https://www.my-mooc.com/en/mooc/intro-to-devops--ud611/>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

CCE 5203: Cloud Security Essentials

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Demonstrate the use of identity and access management features of AWS.

CO2: Demonstrate the available methods for securing data and infrastructure.

CO3: Examine logging and monitoring services to identify the incidence.

Unit I

11 hrs

Introduction to Cloud Security: Introduction, Background and need of Cloud Security. Security design principles. Shared Responsibility models of Cloud.

Threat Modelling Security Fundamentals: Introduction. Create threat model using data flow diagram. Provide context with right depth layer. Approach your data flow diagram with right threat model focus. Use a framework to identify threats and find ways to eliminate or reduce your risks. Prioritize your issues and apply security controls. Use recommendation tools to create data flow diagram.

Securing Access to Cloud Resources: Introduction to Securing Access to Cloud Resources. IAM fundamentals. Authenticating with IAM. Authorizing with IAM. Examples of authorizing with IAM. Additional authentication and access management services. Using AWS Organizations. (MOOCs: Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy)

Unit II

13 hrs

Securing your Infrastructure: Structure of a three-tier web application using a VPC. Setting up public and private subnets and internet protocols. Using AWS security groups. Using AWS network ACLs. Using AWS load balancers. Pulling it all together. Protecting your computer resources.

Protecting Data in your Application: Protect data at rest. Amazon S3 protection features. Protection through encryption. Protect data in transit. Best practices to protect data in Amazon S3. Additional data protection services.

Unit III

12 hrs

Configure and manage SQL database security: Configure which users and applications have access to your SQL databases. Block access to your servers using firewalls. Discover, classify, and audit the use of your data. Encrypt and protect your data while it is stored in the database.

Logging and Monitoring: Importance of logging and monitoring. Capture and collect. AWS services with built-in logs. Monitor and report. Best practices for logging and monitoring. Additional AWS services for logging and monitoring

Responding to and Managing an Incident: Identifying an incident. AWS services that support the discovery and recognition phase. AWS services that support the resolution and recovery phase. Best



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

practices for handling an incident. (MOOCS: Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy).

References

1. Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy
2. Multimedia resource from Microsoft Learn: <https://learn.microsoft.com/en-us/training/modules/sql-database-security/>
3. Multimedia resource from Microsoft Learn: <https://learn.microsoft.com/en-us/training/paths/tm-threat-modeling-fundamentals/>

CYE 5202: Web Security and Secure Coding

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Analyse common web application threats, vulnerabilities, and threat modeling strategies to assess risk and propose appropriate counter measures.

CO2: Apply secure coding practices to mitigate client-side attacks by implementing appropriate counter measures.

CO3: Evaluate API vulnerabilities and implement security measures.

Unit I

10 hrs

Hacking Web Applications: Basic Web Security Model- Web Application Threats & Vulnerabilities (OWASP Top 10) - Objectives of Web Application Hacking - Anatomy of an Attack - Search Engine Reconnaissance & Google Dorking - Modern Attack Techniques - Countermeasures. Web Threat Modeling, Threat Modeling Strategies - CVSS Scoring & Risk Analysis.

Unit II

17 hrs

Client-Side Attacks, Defenses & Secure Coding: Authentication & Password Security - Types of Authentication & Password Cracking - Password Attack - Modern Authentication Methods - Countermeasures. Session Hijacking & Secure Sessions - Application-Level & Network-Level Attacks - Secure Session Management - Tools & Countermeasures. Injection Attacks & Secure Coding Practices - SQL Injection - Secure Coding. Memory Exploits & Buffer Overflows Stack-Based Overflows - Secure Coding Memory-Safe Languages. Cross-Site Scripting (XSS) & CSRF - DOM-Based XSS, Stored & Reflected XSS - CSP, Input Sanitization, CSRF Tokens - Secure Coding.

Web Application Infrastructure Security: Web Server Security - Web Server Exploits - Secure Configuration of Web Servers & Web Apps



Unit III

9 hrs

API Security & Secure Coding Practices: API Security- Common API Vulnerabilities - OWASP API Security Top 10 (2021) - API Authentication Methods - API Security Headers & Rate Limiting Techniques - Introduction to Server-Side Request Forgery & API Hardening - Access Control - API Security Testing

References

1. The Web Application Hacker's Handbook: Discovering and Exploiting Security Flaws, Dafydd Stuttard and Marcus Pinto, 2nd edition, Wiley, 2011.
2. Web Security: A WhiteHat Perspective, Hanqing Wu and Liz Zhao, Auerbach Publications; 1st edition, 2015.
3. Threat modeling designing for security, Adam Shostack, Wiley Publication.
4. Secure Programming Cookbook, J. Viega, M. Messier, O'Reilly, 2003.
5. Writing Secure Code, M. Howard, D. LeBlanc, Microsoft, Second edition, 2002.
6. Building Secure Software, J. Viega, G. McGraw, Addison Wesley, 2002.

ESE 5234: Entrepreneurship

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Explain the importance of entrepreneurship and entrepreneurial development model, social responsibilities of business.

CO2: Describe Entrepreneurial Traits and Factors affecting Entrepreneurship process.

CO3: Discuss Business Start-up Process.

CO4: Summarize a business and marketing plan for entrepreneurs.

Unit I

6 hrs

Introduction to Entrepreneurship: Meaning and Definition of Entrepreneurship-Employment vs Entrepreneurship, Theories of Entrepreneurship, approach to entrepreneurship, Entrepreneurs VS Manager.

Unit II

5 hrs

Entrepreneurial Traits: Personality of an entrepreneur, Types of Entrepreneurs.

Unit III

6 hrs

Process of Entrepreneurship: Factors affecting Entrepreneurship process.

Unit IV

7 hrs

Business Start-up Process: Idea Generation, Scanning the Environment, Macro and Micro analysis.



Unit V

6 hrs

Business Plan writing: Points to be considered, Model Business plan.

Unit VI

6 hrs

Case studies: Indian and International Entrepreneurship.

References

1. NVR Naidu and T. Krishna Rao, "Management and Entrepreneurship", IK International Publishing House Pvt. Ltd 2008.
2. Mohanthy Sangram Keshari, "Fundamentals of Entrepreneurship", PHI Publications, 2005
3. <https://nptel.ac.in/courses/110106141>

Elective - II (Theory)

AME 5003: Principles of Natural Language Processing

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Understand text processing for information extraction and retrieval.

CO2: Compare and contrast state-of-the-art word vectorization models for real-world problems.

CO3: Apply deep learning-based approaches for natural language processing tasks.

Unit I

10 hrs

Introduction to Text Processing: Information extraction using regular expressions - Stages of text processing: tokenization, normalization, stemming, lemmetization and stop words-removal.

Basic linguistic processing: Part-of-Speech (POS) tagging and Named Entity Recognition (NER).

Information Retrieval: term document incidence matrices, inverted index construction and query processing, basic ranking using TF-IDF.

Unit II

12 hrs

Language Modelling, Sentiment Analysis, Word Embeddings: Language modelling: N-grams, N-gram probabilities, evaluation and perplexity. Sentiment Analysis using Naïve Bayes.

Text representation models: Bag-of-Words, TF-IDF vector model, cosine similarity. Word embeddings: Neural word embeddings - Word2Vec (CBOW and Skip-gram) and GloVe models

Unit III

14 hrs

NLP with Deep Learning, Applications of Natural Language Processing: Neural approaches for NLP: Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for sequence modelling, GRU (Gated Recurrent Unit) Attention mechanism and sequence-to-sequence models.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Transformer architecture: Encoder and decoder transformers, self-attention mechanism.

Pre-trained language models: BERT and DitiBERT fine-tuning for downstream NLP tasks.

Applications of NLP: Machine translation, question answering, chatbots, and text summarization.

References

1. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
2. S. Vajjala, B. Majumder, A. Gupta, and H. Surana, *Practical Natural Language Processing*. Sebastopol, CA, USA: O'Reilly Media, Jun. 2020.
3. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (draft). [Online]. Available: <https://web.stanford.edu/~jurafsky/slp3/>
4. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*, 1st ed. London, U.K.: ISTE Ltd., 2017. [Online]. Available: <https://www.nltk.org/book/>
5. Y. Goldberg, *A Primer on Neural Network Models for Natural Language Processing*. [Online]. Available: <http://faculty.cse.tamu.edu/huangrh/Spring18/nnlp.pdf>
6. D. Rao and B. McMahan, *Natural Language Processing with PyTorch*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2019.

AME 5205: Advanced Generative AI and Agentic Systems

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Design knowledge-augmented generative AI systems using retrieval and embedding models.

CO2: Evaluate advanced large language model architectures and parameter-efficient fine-tuning techniques.

CO3: Design agentic AI systems and assess reliability, safety, and ethical considerations in generative AI applications.

Unit I

10 Hrs

Large Language Models (LLMs): Large Language Models (LLM), Prompt Engineering, Limitations of LLM, Knowledge-augmented generative AI approaches: Retrieval Augmented Generation (RAG) architectures and modular RAG pipeline design. Dense retrieval methods and embedding-based search. Document chunking strategies and semantic indexing.

Hallucination in generative AI: causes, detection techniques, and evaluation of faithfulness and groundedness.

Applications of retrieval-based generative AI systems.



Unit II

12 Hrs

Advanced Large Language Model Architectures: Large Language Model architectures: GPT architecture and comparison of modern open and closed LLM architectures. Long context handling techniques, Instruction tuning and alignment methods including Reinforcement Learning from Human Feedback (RLHF), Reinforcement Learning from AI Feedback (RLAIF).

Parameter-efficient fine-tuning techniques including LoRA and QLoRA.

Unit III

14 Hrs

Agentic AI and Autonomous Generative Systems: Design principles of agentic AI, Tree-of-Thought reasoning, Memory architectures in AI agents, Multimodal agent systems, Speech-enabled conversational agents.

Reliability and safety in agentic systems: Agent failure modes, Self-correction and verification loops, Safety guardrails

Responsible, Secure Generative AI Systems: Ethical and responsible generative AI, Bias and fairness in foundation models, Explainability challenges in generative AI, Responsible AI frameworks, Governance and regulatory considerations.

References

1. D. Foster, Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.
2. C. M. Bishop and H. Bishop, Deep Learning: Foundations and Concepts. Cham, Switzerland: Springer, 2024.
3. J. Alammam and M. Grootendorst, Hands-On Large Language Models: Language Understanding and Generation. Sebastopol, CA, USA: O'Reilly Media, 2024.
4. Hugging Face, "Transformers Documentation." [Online]. Available: <https://huggingface.co/docs/transformers>
5. Hugging Face, "PEFT: Parameter Efficient Fine-Tuning." [Online]. Available: <https://huggingface.co/docs/peft>
6. DeepLearning.AI, "Generative AI for Everyone," Coursera. [Online]. Available: <https://www.coursera.org/learn/generative-ai-for-everyone>
7. TransformaTech Institute, Creating Production-Ready LLMs: Building, Optimizing, and Deploying Large Language Models, 2024.

CCE 5241: Cloud Operations

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Understand the architectural decisions and use cloud services to build decoupled serverless microservices.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

CO2: Analyse the tools specific to cloud operations engineer to create reusable compute and networking cloud infrastructures.

CO3: Apply cloud architecting solutions to manage, automate, secure, and monitor highly available and fault tolerant infrastructures.

Unit I

10 hrs

Building Decoupled Architecture: Architectural need, Decoupling your architecture, Decoupling with Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS), Sending messages between cloud applications and on-premises with Amazon MQ.

Building Microservices and Serverless Architectures: Architectural need, Introducing microservices, building microservice applications with AWS container services, Deploy Docker Container on Amazon Elastic Container Service (ECS), Introducing serverless architectures, building serverless architectures with AWS Lambda, Extending serverless architectures with Amazon API Gateway, Orchestrating microservices with AWS Step Functions.

Caching Content: Architectural need, Architectural need, Overview of caching, Edge caching, Caching web sessions, Caching databases, Streaming Dynamic Content Using Amazon Cloud Front.

Planning for Disaster: Architectural need, Disaster planning strategies, Disaster recover patterns, Hybrid Storage and Data Migration with AWS Storage Gateway, File Gateway.

Unit II

14 hrs

Understanding Systems Operations on AWS:

Systems Operations in the Cloud, Introduction to Core Services, AWS Identity and Access Management (IAM), IAM Policy Simulator, AWS Command Line Interface (CLI) Educator Demo AWS CLI.

Tooling and Automation: AWS Systems Manager, Additional Administration and Development Tools, Hosting a Static Website on Amazon S3. (MOOCS: Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy).

Scaling and Name Resolution: Architectural need, Elastic Load Balancing, Amazon EC2 Auto Scaling, Amazon Route 53, Amazon Route 53 Routing Options Demo, Amazon Route 53 Geo Routing Demo, Configure Failover Routing with Amazon Route 53.

Unit III

12 hrs

Monitoring and Security: Amazon CloudWatch - Monitoring, Events, Logs, AWS CloudTrail, WS Service Integration with Amazon, Athena, AWS Configre Amazon CloudWatch, Amazon CloudWatch Monitoring, Amazon CloudWatch Events, Amazon CloudWatch Logs, AWS CloudTrail, AWS Service Integration with Amazon Athena, AWS Config.

Managing Resource Consumption: Tagging, Cost Management Tools and Best Practices, AWS Trusted Advisor, Optimize AWS Resource Utilization.

(MOOCS: Multimedia resource from AWS Academy: [aws.amazon.com/training/ awsacademy](https://aws.amazon.com/training/awsacademy)).



Cloud Architecting: Configuration Management in the Cloud, Using Configuration Software, Creating an AMI Building Strategy, Amazon EC2 Launch Templates, Infrastructure as Code, Introduction to JSON and YAML, AWS Cloud Formation, Troubleshooting AWS Cloud Formation, Continuous Integration and Continuous Deployment on AWS.

Reference

1. Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy

CYE 5203: Machine Learning for Cybersecurity

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Demonstrate various data analytics and data visualization techniques for cybersecurity applications.

CO2: Implement and assess various machine learning models to detect and mitigate security threats.

CO3: Analyse adversarial machine learning techniques.

Unit I

15 hrs

Foundations of Machine Learning: Overview of Supervised (regression and classification), unsupervised (clustering and dimensionality reduction), semi-supervised, and reinforcement learning with practical examples - Machine learning nomenclature: raw data, types of features and outputs, feature vector, Handling missing data, encoding categorical variables, feature scaling and normalization.

Classification and regression using decision trees, splitting criteria, entropy, information gain, Gini impurity - Overfitting & Pruning in decision trees.

Linear regression: Model, metrics (MSE, RMSE, R-square), estimation, and interpretation of coefficients, Introduction to bias/variance trade-off, regularization methods (Ridge and Lasso).

Unit II

15 hrs

Logistic regression for binary classification: Sigmoid function, decision boundary, model interpretation. Model evaluation and validation: Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.

Nearest neighbours' algorithms - Probabilistic modelling: Nearest neighbours' algorithms - Probabilistic modelling of data using parameters, Introduction to maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification.

Ensemble learning methods: Bagging and Random Forest, Boosting techniques. Introduction to hyperparameter tuning and model improvement.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

K-means clustering - computing distances and similarities, limitations, Hierarchical clustering-dendrogram construction, types of linkage, Dimension reduction using principal component analysis (PCA)-concept, mathematics and applications.

Neural networks: Multilayer perceptron (MLP), Backpropagation algorithm and learning intuition, Overfitting and regularization in NNs.

Unit III

6 hrs

Adversarial Machine Learning & Attacker Techniques: Privacy Breaches and Data Poisoning, Adversarial Attacks on ML Models, Model Stealing and Evasion Techniques, principle of autoencoders and GANs for threat and anomaly detection.

References

1. Network Anomaly Detection: A Machine Learning Perspective, Dhruba K. Bhattacharyya, 1st Edition, Chapman and Hall/CRC; 2013.
2. Machine Learning for Hackers: Case Studies and Algorithms to Get You Started, Drew Conway 1st Edition, Shroff, 2012
3. Malware Data Science by Joshua Saxe and Hillary Sanders. Learning for Audio, Image and Video Analysis - Theory and Applications, Springer, No Starch Press, 2018.
4. Module: Generative AI with Large Language Models
(<https://www.coursera.org/learn/generative-ai-with-llms>)

ESE 5244: IT Project Management

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Illustrate the importance of project planning.

CO2: Discuss and demonstrate various tools applicable for different phases of the software project.

CO3: Illustrate the importance of Change management.

CO4: Illustrate the importance of team work and demonstrate the usefulness of tools in effective handling of the project.

CO5: Compare the problems in projects that don't follow the process and discuss new trends in software life cycle.

Unit I

3 hrs

Software Project Planning: Understand the Project Needs, Create the Project Plan, Diagnosing Project Planning Problems.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Unit II

3 hrs

Estimation: Elements of a Successful Estimate, Wideband Delphi Estimation, Other Estimation Techniques, Diagnosing Estimation Problems.

Unit III

3 hrs

Project Schedules: Building the Project Schedule, Managing Multiple Projects, Use the Schedule to Manage Commitments, Diagnosing Scheduling Problems.

Unit IV

4 hrs

Reviews: Inspections, Deskchecks, Walkthroughs, Code Reviews, Pair Programming, Use Inspections to Manage Commitments, Diagnosing Review Problems.

Unit V

4 hrs

Software Requirements: Requirements Elicitation, Use Cases, Software Requirements Specification, Change Control, Introduce Software Requirements Carefully, Diagnosing Software Requirements Problems.

Unit VI

4 hrs

Design and Programming: Review the Design, Version Control with Subversion, Refactoring, Unit Testing, Use Automation, Be Careful with Existing Projects, Diagnosing Design and Programming Problems.

Unit VII

4 hrs

Software Testing: Test Plans and Test Cases, Test Execution, Defect Tracking and Triage, Test Environment and Performance Testing, Smoke Tests, Test Automation, Postmortem Reports, Using Software Testing Effectively, Diagnosing Software Testing Problems.

Unit VIII

3 hrs

Understanding Change: Why Change Fails, How to Make Change Succeed.

Unit IX

12 hrs

Management and Leadership: Take Responsibility, Do Everything Out in the Open, Manage the Organization, Manage Your Team.

Unit X

3 hrs

Managing an Outsourced Project: Prevent Major Sources of Project Failure, Management Issues in Outsourced Projects, Collaborate with the Vendor.



Unit XI

3 hrs

Process Improvement: Life Without a Software Process, Software Process Improvement, Moving Forward.

References

1. “Applied Software Project Management”, Jennifer Greene, Andrew Stellman (O'Reilly Publications), 2005.
2. “The Art of Project Management”, Scott Berkun (O'Reilly Publications), 2005.

AME 5252: Deep Learning Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply state-of-the-art deep learning tools and libraries.

CO2: Implement and investigate techniques for improving the way neural networks learn.

CO3: Apply and analyze deep learning models for real-world problems.

Unit I

12 hrs

Deep Neural Networks: Visualize different nonlinear activation functions; Implement forward and backward propagation for a shallow neural network efficiently using a matrix-based approach; Implement gradient descent method for a shallow neural network; Implement different loss functions and study their effectiveness for real-world problems; Numerically investigate the learning slowdown problem in deep neural networks: backward propagating gradient shrinkage due to softmax and activation function gradients, vanishing and exploding gradients; Numerically implement and investigate the effect of weight initialization, L1/L2 & dropout regularizations, residual & skip connections, batch normalization, adaptive learning, learning rate scheduling, gradient clipping, early stopping, and data augmentation.

Unit II

12 hrs

Transformer Networks: Implement and understand word embeddings from language models; Implement and understand single and multi-head attention mechanisms; Implement different types of transformers such as vision transformers, generative image transformers, and multimodal transformers for real-world problems.

Unit III

12 hrs

Advanced Topics in Deep Learning: Implement and understand transfer Learning from existing state-of-the-art model; Implement Generative adversarial networks and autoencoders for real-world problems such as anomaly detection; Develop skills in building and implementing problem-specific small-scale models using knowledge distillation and quantization.



References

1. Neural Networks and Deep Learning, Michael Nielsen - Determination Press - Available online at <http://neuralnetworksanddeeplearning.com/index.html>
2. Deep Learning with Python, François Chollet, Manning Publications; 2nd Edition, 2021.
3. Deep Learning Foundations and Concepts, Christopher M. Bishop and Hugh Bishop, Springer; 2024.
4. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
5. Lecture slides of Prof. Andrew Ng - Stanford University - Available online at <https://cs230.stanford.edu/syllabus/>
6. Tensor Flow for Deep Learning Paperback, Reza Zadeh & Bharath Ramsundar, O'Reilly, 2018.
7. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville - MIT Press - Available online at <http://www.deeplearningbook.org/>
8. Learning path: Microsoft Azure AI Fundamentals: Explore natural language processing (<https://learn.microsoft.com/en-us/training/paths/explore-natural-language-processing/>).

BDE 5252: Modern Databases for Big Data Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply the principles of relational database management.

CO2: Apply and evaluate different transaction processing and lock management methods.

CO3: Apply the principles of document DB and vector DB.

Unit I

12 hrs

Relational database - import and manipulate data.

Unit II

12 hrs

Transaction processing and lock management.

Unit III

12 hrs

Document DB and vector DB.

References

1. Machine Learning for Audio, Image and Video Analysis, Francesco Camastra and Alessandro Vinciarelli Springer's Publication, 2nd edition. 2015.
2. Practical Python and OpenCV, An Introductory, Example Driven Guide to Image Processing and Computer Vision, Dr. Adrian Rosebrock, 4th edition, 2019



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

3. Computer Vision with Python Cookbook: Leverage the power of OpenCV 3 and Python to build computer vision applications, Aleksei Spizhevoi, Aleksandr Rybnikov, Packt Publishing, 1st Edition, 2018.

ESE 5251: Device Drivers Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Demonstrate the development of basic modules and drivers.

CO2: Examine the ways of debugging and handling concurrency in device drivers.

CO3: Examine the implementation of interrupts and work queues for deferred work.

Unit I

12 hrs

Introduction to device drivers & running module.

Unit II

12 hrs

Character driver.

Debugging techniques, concurrency and race condition.

Unit III

12 hrs

Advanced character driver operations, time, delay and deferred work, allocating Memory

Communicating with Hardware, Interrupt Handling, PCI Drivers, USB Drivers.

References

1. Alessandro Rubini, "Linux Device Drivers", (Nutshell Handbook), O'Reilly Publishers, 2009.
2. John Madiou, "Linux Device Drivers Development: Develop customized drivers for embedded Linux", Packt Publishing, 2017.
3. Robert Love, "Linux Kernel Development", Addison Wesley, Third Edition, 2010.
4. Daniel P. Bovet, Marco Cesati, "Understanding the Linux Kernel", O'Reilly Media, Third Edition, 2008.
5. Wolfgang Mauerer, "Professional Linux Kernel Architecture", Wrox, 2008.
6. Sreekrishnan Venkateswaran, "Essential Linux Device Drivers", Prentice Hall, 2008.
7. W. Richard Stevens, Stephen A. Rago, "Advanced Programming in the UNIX Environment", Addison Wesley, Third Edition, 2013.
8. W. Richard Stevens, Bill Fenner, Andrew M. Rudoff, "Unix Network Programming, Vol1: Sockets", Pearson Education India, Third Edition, 2015.



Elective - I (Lab)

AME 5254: Applied Computer Vision Lab

(L-T-P-C: 0-0-3-1)

Total 36 Hours

Course Outcome

CO1: Implement transfer learning for vision tasks

CO2: Apply detection and segmentation workflows

CO3: Deploy vision models for real-time inference

Unit I

10 hours

Transfer Learning for Vision: Implementation of transfer learning for image classification using pretrained CNN models including data augmentation and model evaluation.

Unit II

14 hours

Detection & Segmentation: Object detection and segmentation workflows using deep vision models with performance benchmarking.

Unit III

12 hours

Vision Deployment: Real-time inference, latency benchmarking, edge deployment, API-based deployment, and monitoring of vision model performance.

References

1. Learning path: Microsoft Azure AI Fundamentals: Explore computer vision (<https://learn.microsoft.com/en-in/training/paths/explore-computer-vision-microsoft-azure/>).
2. Generative AI for Everyone from DeepLearning.AI (<https://www.coursera.org/learn/generative-ai-for-everyone>).
3. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
4. Lecture slides of Prof. Fei-Fei Li - Stanford University - Available online at <http://cs231n.stanford.edu/>
5. A Guide to Convolutional Neural Networks for Computer Vision, Salman Khan, Hossein Rahmani, Syed Afaq Ali Shah, and Mohammed Bennamoun, Morgan & Claypool Publishers, 2018.
6. Lecture slides of Prof. Fei-Fei Li - Stanford University - Available online at <http://cs231n.stanford.edu/>
7. Neural Networks and Deep Learning, Michael Nielsen, Determination Press - Available online at <http://neuralnetworksanddeeplearning.com/index.html>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

8. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2011 - Online resource from Springer available at <http://szeliski.org/Book/>

BDE 5253: Data Engineering Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Identify fundamental concepts of data engineering ecosystem components.

CO2: Design data engineering pipelines using data lake principles and Medallion architecture.

CO3: Develop automated data engineering workflows using orchestrators.

Unit I

9 hrs

Setting up the data engineering environment.

Data ingestion exercises: Collecting data from APIs, files, and databases.

Building a simple data pipeline workflow consisting of ingestion, transformation, and serving layers.

Unit II

12 hrs

Creating and managing datasets in different storage systems such as local file systems and object storage.

Implementing data lake concepts by storing raw datasets and performing cleaning and transformation tasks.

Designing and implementing Medallion Architecture (Bronze, Silver, Gold layers) for a sample dataset.

Unit III

15 hrs

Creating simple Directed Acyclic Graphs (DAGs) to automate data pipeline tasks.

Implementing Airflow operators to perform data ingestion, transformation, and aggregation tasks.

Implementing task dependencies, retries, scheduling, and logging mechanisms in Airflow pipelines.

References

1. J. Reis and M. Housley, *Fundamentals of Data Engineering: Plan and Build Robust Data Systems*. Sebastopol, CA, USA: O'Reilly Media, 2022.
2. M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*. Sebastopol, CA, USA: O'Reilly Media, 2017.
3. T. Akidau, S. Chernyak, and R. Lax, *Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing*. Sebastopol, CA, USA: O'Reilly Media, 2018.
4. M. Armbrust *et al.*, "Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics," in *Proc. Conf. Innovative Data Systems Research (CIDR)*, 2021.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

5. Apache Software Foundation, "Apache Airflow Documentation," 2023. [Online]. Available: <https://airflow.apache.org>

CCE 5051: DevOps for Cloud Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

- CO1: Implement Linux and version control system commands in DevOps Life Cycle stages.
CO2: Design CI/CD pipeline to automate software product development lifecycle.
CO3: Design configuration management and automation templates to provision infrastructure and configure software in DevOps lifecycle.

Unit I

12 hrs

Understand the lifecycle stages of DevOps product life cycle methodology.
Explore the basic Linux commands, system utilities and services.
Understand the basic architecture of software version controls like GIT, SVN.
Setup a public git repo for an application, explore the GIT commands to perform clone, fork, pull, push, commit etc.

Unit II

15 hrs

Demonstrate a continuous integration pipeline using Jenkins master and slave architecture for web applications
Explore containerizing web applications using Docker containers
Setup a Kubernetes cluster to deploy containerised application with high availability and load balancing.

Unit III

9 hrs

Create Ansible playbook for configuring Linux and window host.
Setup a continuous monitoring service for managing IT assets.

References

1. Beginning Shell Scripting (Programmer to Programmer), Eric Foster-Johnson, John C. Welch, Micah Anderson, Wrox Publications, 2005.
2. Mastering Unix Shell Scripting: Bash, Bourne, and Korn Shell Scripting for Programmers, System Administrators, and UNIX Gurus, Randal K. Michael, 2nd Edition, Wiley Publications 2nd edition, 2008.
3. UNIX & Shell Programming, Bintu Harwani, Oxford Publications, 2013.
4. Jenkins: The Definitive Guide, John Ferguson Smart, O'Reilly Media, Inc., 2011.
5. Jenkins Essentials, Mitesh Soni, Packt Publications, 2015.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

6. Continuous Delivery with Docker and Jenkins, Rafal Leszko, Packt Publications, Second Edition, 2019.
7. Implementing DevOps on AWS, Veselin Kantse, Packt Publications, 2017
8. Docker Orchestration, Randall Smith, Packt Publications, 2017.
9. Jenkins Continuous Integration Cookbook, Alan Berg, Packt Publications, 2012.
10. Practical LXC and LXD Linux Containers for Virtualization and Orchestration, Kumaran S., Senthil, Apress Publications, 2017.
11. Containerization with LXC, Konstantin Ivanov, Packt Publications, Packt Publishing; 1st edition, 2017.
12. Docker: Up & Running: Shipping Reliable Containers in Production, Karl Matthias, Sean Kane, O'Reilly Media. O'Reilly Media; 1st edition, 2015.

CCE 5253: Cloud Security Essentials Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Experiment the features of identity and access management services in AWS.

CO2: Design a setup for securing data and infrastructure.

CO3: Develop an architecture for handling monitoring and incident response using AWS services.

Unit I

12 hrs

Introduction to Cloud Security: Threat Modelling Security Fundamentals, Securing Access to Cloud Resources.

Unit II

12 hrs

Securing your Infrastructure: Protecting Data in your Application.

Unit III

12 hrs

Configure and manage SQL database security: Logging and Monitoring, Responding to and Managing an Incident.

References

1. Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy.
2. Multimedia resource from Microsoft Learn: <https://learn.microsoft.com/en-us/training/modules/sql-database-security/>
3. Multimedia resource from Microsoft Learn: <https://learn.microsoft.com/en-us/training/paths/tm-threat-modeling-fundamentals/>



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

CYE 5252: Web Security and Secure Coding Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply reconnaissance, vulnerability scanning, and exploit web application security flaws using tools like Burp Suite, OWASP ZAP, sqlmap, and Hydra.

CO2: Apply secure coding techniques to mitigate security vulnerabilities, including SQL Injection, XSS, CSRF, authentication flaws, session hijacking, and buffer overflow.

CO3: Analyse threat modeling and risk assessment for web applications.

Unit I

12 hrs

Web Application Reconnaissance & Google Dorking: Perform Google Dorking to extract sensitive information - Use Shodan to identify exposed web services.

Web Vulnerability Scanning & OWASP Top 10 Analysis: Set up OWASP Juice Shop/Damn Vulnerable Web Application (DVWA) - Perform automated web vulnerability scanning using OWASP ZAP/Nikto - Analyze OWASP Top 10 vulnerabilities in a real-world application.

Authentication & Password Security: Test weak password authentication using Hydra, John the Ripper, and Hashcat. - Analyze session security flaws using Burp Suite.

Unit II

18 hrs

Session Hijacking & Secure Session Management: Capture and manipulate session cookies using Burp Suite - Perform session fixation and session hijacking attacks - Implement secure session management techniques.

SQL Injection Attacks & Secure Coding: Perform SQL injection attacks on DVWA and Juice Shop. - Use tools like sqlmap for automated SQLi exploitation - Implement prepared statements and input validation to mitigate SQLi.

Buffer Overflow & Memory Exploits: Exploit a stack-based buffer overflow in a C program - Analyze memory vulnerabilities using GDB and Pwntools - Apply secure coding practices to prevent memory corruption.

Cross-Site Scripting (XSS) & Cross-Site Request Forgery (CSRF): API Security & OWASP API Top 10 - Web Server Security & Secure Configurations.

Unit III

6 hrs

Web Threat Modeling & Risk Analysis - STRIDE, Attack Tree - CVSS.

References

1. Web Security: A WhiteHat Perspective, Hanqing Wu and Liz Zhao, Auerbach Publications; 1st edition, 2015.
2. Threat modeling designing for security, Adam Shostack, Wiley Publication.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

3. Writing Secure Code, Michael Howard and David LeBlanc, Microsoft Press, 2nd Edition, 2004
4. Buffer Overflow Attacks: Detect, Exploit, Prevent by Jason Deckard, Syngress, 1st Edition, 2005.
5. Foundations of Security. Neil Daswani, Christoph Kern, and Anita Kesavan. Apress, 2007 (1st ed). ISBN-10: 1590597842; ISBN-13: 978-1590597842. This book is accessible online from machines on the USF network: <https://link.springer.com/book/10.1007/978-1-4302-0377-3>

ESE 5284: Entrepreneurship Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Understand prominence of entrepreneurship.

CO2: Develop use cases for building a business.

CO3: Evaluation of factors influencing business venture.

Unit I

6 hrs

Study of use cases for need and prominence of entrepreneurship, associated decision making process.

Unit II

6 hrs

Study of report by the National Knowledge Commission on the importance of entrepreneurship in economic development.

Unit III

9 hrs

Develop use cases for identifying and evaluating opportunities, developing business plan, assessment of resources, project appraisal and feasibility plan.

Unit IV

9 hrs

Creating and starting venture includes legal requirements, marketing strategies, financial plans and human resources management

Unit V

6 hrs

Design a Case studies of Indian and International Entrepreneurship.

References

1. Management and Entrepreneurship, NVR Naidu and T. Krishna Rao, IK International Publishing House Pvt.Ltd, 2008.
2. Fundamentals of Entrepreneurship, Mohanthy Sangram Keshari, PHI Learning Pvt. Ltd., 2005.



Elective - II (Lab)

AME 5053: Principles of Natural Language Processing Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement text processing techniques for information extraction & retrieval.

CO2: Apply state-of-the-art word vectorization models for real-world problems.

CO3: Formulate and apply deep learning-based approaches for natural language processing tasks.

Unit I

9 hrs

Introduction to Text Processing: Implement information extraction using regular expressions, implement text processing tasks such as tokenization, normalization, stemming, and stop words-removal, implement information retrieval using term document incidence matrices, implement inverted index construction and query processing.

Unit II

12 hrs

Language Modelling, Sentiment Analysis, Word Embeddings: Implement n-grams and n-gram probabilities, implement sentiment Analysis using Naïve Bayes, implement cosine similarity and understand its significance in NLP, implement word vectorization models such as TF-IDF, Word2Vec, & GloVe, and compare them for real-world applications, Visualize word embeddings.

Unit III

15 hrs

NLP with Deep Learning, Applications of Natural Language Processing: Implement sequence models for NLP tasks using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks. Implement attention-based sequence-to-sequence models for text processing tasks. Implement transformer-based language models such as encoder and decoder transformers.

Apply pre-trained transformer models such as BERT and perform fine-tuning for NLP applications such as text classification, question answering, or text summarization.

References

1. IBM, "AI Engineering Professional Certificate." [Online]. Available: <https://www.coursera.org/professional-certificates/ai-engineer>
2. S. Vajjala, B. Majumder, A. Gupta, and H. Surana, *Practical Natural Language Processing*. Sebastopol, CA, USA: O'Reilly Media, Jun. 2020.
3. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (draft). [Online]. Available: <https://web.stanford.edu/~jurafsky/slp3/>



4. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*, 1st ed. London, U.K.: ISTE Ltd., 2017. [Online]. Available: <https://www.nltk.org/book/>
5. Y. Goldberg, *A Primer on Neural Network Models for Natural Language Processing*. [Online]. Available: <http://faculty.cse.tamu.edu/huangrh/Spring18/nnlp.pdf>
6. D. Rao and B. McMahan, *Natural Language Processing with PyTorch*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2019.

AME 5255: Advanced Generative AI and Agentic Systems Lab

(L-T-P-C: 0-0-3-1)

Total 36 Hours

Course Outcome

CO1: Implement retrieval-augmented generative AI systems using embedding models and vector databases.

CO2: Apply prompt engineering and parameter-efficient fine-tuning techniques to adapt large language models.

CO3: Design simple agentic AI workflows and evaluate reliability, safety, and ethical aspects of generative AI systems.

Unit I

10 hrs

Introduction to generative AI tools: prompt engineering experiments, exploring limitations and hallucinations in large language models, generating text using models from Hugging Face Transformers, creating text embeddings using SentenceTransformers, semantic similarity search using vector databases such as FAISS, document chunking and indexing strategies, implementation of a basic Retrieval-Augmented Generation (RAG) pipeline using LangChain, evaluation of RAG outputs for faithfulness and hallucination reduction.

Unit II

12 hrs

Generative Learning Algorithms: Exploration of large language model architectures, instruction-based prompting techniques, implementing prompt templates and prompt chaining, demonstration of instruction tuning workflows, parameter-efficient fine-tuning using LoRA, fine-tuning using the PEFT library, comparison of base model responses versus adapted model responses, evaluating model outputs for accuracy and relevance.

Unit III

14 hrs

Design of simple AI agents using LangChain: implementation of multi-step reasoning using Tree-of-Thought prompting, building conversational agents with memory modules, designing speech-enabled conversational agents using APIs, monitoring AI agent responses and failure modes, implementing guardrails and safety filters, evaluating bias and fairness in generated outputs, basic explainability and responsible AI considerations in generative systems.



References

1. L. Wu et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, arXiv, 2020.
2. C. Huyen, *AI Engineering: Building Applications with Foundation Models*. Sebastopol, CA, USA: O'Reilly Media, 2024.
3. S. Raschka, *Build a Large Language Model from Scratch*. Birmingham, U.K.: Packt Publishing, 2024.
4. T. Tunstall, L. von Werra and T. Wolf, *Natural Language Processing with Transformers*. Sebastopol, CA, USA: O'Reilly Media, 2022.
5. Hugging Face, *Transformers Documentation*. [Online]. Available: <https://huggingface.co/docs>

CCE 5291: Cloud Operations Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Examine the architectural decisions and use cloud services to build decoupled serverless microservices.

CO2: Experiment the tools specific to cloud operations engineer to create reusable compute and networking cloud infrastructures.

CO3: Design cloud architecting solutions to manage, automate, secure, and monitor highly available and fault tolerant infrastructures.

Unit I

12 hrs

Demonstrate a serverless architecture for web application using services like AWS SQS, Lambda and SNS. Deploy a microservices using Elastic Container Service (ECS). Setup content delivery network to distribute static web content across different regions of globe using AWS global Infrastructure. Implement high available system with disaster recover strategies.

Unit II

12 hrs

Explore AWS CLI for performing administrative operations on AWS services. Explore different features of AWS system manager to perform operations on EC2 Instance. Deploy a Static Website using S3 Bucket. Design a highly available, fault-tolerant and scalable dynamic architecture for web application using core Aws services.

Unit III

12 hrs

Create a Cloud Watch alarm and events to monitor performance of AWS resources. Explore AWS Cloud Watch trail and logs. Understand best practices for designing a well architected framework. Illustrate



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

how tagging a resource helps to optimise the cost. Create a Cloud Formation template to deploy network and application layer of web application across different regions of AWS infrastructure.

Reference

1. Multimedia resource from AWS Academy: aws.amazon.com/training/awsacademy

CYE 5253: Machine Learning for Cybersecurity Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply data preprocessing, feature extraction, and visualization to detect cybersecurity threats.

CO2: Implement and evaluate machine learning algorithms for intrusion detection and anomaly detection.

CO3: Understand adversarial machine learning techniques and apply counter measures using AI-based security strategies.

Unit I

12 hrs

Introduction to Data Analytics: Data Cleaning & Preprocessing, Feature Engineering for Cybersecurity.

Visualization Techniques for Security Data: Data Visualization for Threat Detection, Clustering for Anomaly Detection.

Unit II

18 hrs

Introduction to Machine Learning: Supervised vs. Unsupervised Learning for Security, Machine Learning Algorithms for Security, Decision Trees for Cyber security, Ensemble Learning for Threat Detection, Dimensionality Reduction for Security Data, Handling Imbalanced Data in Cyber security.

Unit III

6 hrs

Adversarial Machine Learning & Attacker Techniques: Adversarial Attack on ML Models, Detecting and Mitigating Adversarial Attacks.

References

1. Network Anomaly Detection: A Machine Learning Perspective, Dhruva K. Bhattacharyya, 1st Edition, Chapman and Hall/CRC, 2013.
2. Machine Learning for Hackers: Case Studies and Algorithms to Get You Started, Drew Conway 1st Edition, Shroff, 2012.
3. Malware Data Science by Joshua Saxe and Hillary Sanders. Learning for Audio, Image and Video Analysis - Theory and Applications, Springer, No Starch Press, 2018.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

ESE 5294: IT Project Management Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Identify the skills and techniques required for project life cycle.

CO2: Illustrate essential principles associated with project management and their application in business environment.

CO3: Outline commonly available project management tools.

Unit I

12 hrs

Software project management and its need, evaluation techniques, Project cost estimation techniques.

Unit II

12 hrs

Activity scheduling techniques, cost monitoring, contract management and its necessity risk analysis.

Unit III

12 hrs

Software quality enhancement techniques, people management in software environment and project team structure.

References

1. "Applied Software Project Management", Jennifer Greene, Andrew Stellman (O'Reilly Publications), 2005.
2. "The Art of Project Management", Scott Berkun (O'Reilly Publications), 2005.

MPT 5200: Mini Project - II

(L-T-P-C: 0-0-0-4)

Total 48 hours

Course Outcome

CO1: Identify and formulate a real-world, technologically or socially relevant problem.

CO2: Design, experiment and execute the mini project effectively as a team, using appropriate tools and methodologies.

CO3: Demonstrate the ability to communicate the findings through written reports and oral presentations.

Unit I

6 hrs

Problem identification, literature survey, formation of detailed specifications.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Unit II

30 hrs

Design and implementation of the proposed system.

Unit III

12 hrs

Presentation and demonstration of the work carried out to a panel of experts.

References

1. Research Methodology, Methods and Techniques, C. R. Kothari; New Age International Publishers; 2nd Revised Edition, 2004
(<https://www.jsscacs.edu.in/sites/default/files/Files/Statistics%20and%20Research%20by%20Kothari.pdf>).
2. Project Management by Adrienne Watt - BCcampus Open Textbook project
(https://www.opentextbooks.org.hk/system/files/export/15/15694/pdf/Project_Management_15694.pdf).
3. Research articles and Online Resources.

PSD 5200: Professional Skill Development - II

(L-T-P-C: 0-0-0-1)

Total 12 hours

Course Outcome

CO1: Develop the skills needed for approaching technical and HR interviews.

CO2: Demonstrate the depth of knowledge in the chosen field of study.

CO3: Use mathematical, reasoning, and domain specific skills to solve objective questionnaires in time.

Unit I

4 hrs

Peer interviews, mock interviews.

Unit II

4 hrs

Conduction of domain specific knowledge test using micro-credentials as follows:

Kubernetes:

<https://www.coursera.org/learn/kubernetes-for-absolute-beginners>

(or)

Introduction to Machine Learning: https://onlinecourses.nptel.ac.in/noc24_cs101/preview

Unit III

4 hrs

Analytical and logical reasoning.



References

1. R S Aggarwal. Quantitative Aptitude for Competitive Examinations. S Chand, 2017.
2. McDowell, Gayle Laakmann. Cracking the coding interview: 189 programming questions and solutions. CareerCup, LLC, 2015.
3. Domain specific tools and online resources.

Semesters III & IV

CSE 6098: Project Work

(L-T-P-C: 0-0-0-30)

Total 360 hours

Course Outcome

CO1: Undertake innovative industry/research oriented projects and perform feasibility analysis for finding solutions.

CO2: Implement and test the proposed design using appropriate framework, programming language and tools.

CO3: Demonstrate an ability to present and defend project work carried out to a panel of experts.

Unit I

30 hrs

Problem identification, literature survey, formation of detailed requirement specification document.

Unit II

300 hrs

Design and implementation of the proposed modules with specific test cases.

Unit III

30 hrs

Detailed report of the work carried out, present, and defend the project work carried out to a panel of experts.

Reference

1. Research articles, domain specific tools and online resources.



Program Outcome and Course Outcome: Mapping

Sl. No.	Course Code	Course Name	Credits	PO1	PO2	PO3	PO4	PO5
1	AME 5002	Applied Machine Learning	3	*		*	*	
2	AME 5101	Applied Linear Algebra	3			*	*	
3	CCE 5101	Cloud Application with JAVA and Database	3			*	*	*
4	ESE 5102	Data Structures and Algorithms	3			*	*	*
5	ESE 5104	Real Time Operating Systems	3			*	*	*
6	AME 5052	Applied Machine Learning Lab	1	*		*	*	
7	AME 5151	Applied Linear Algebra Lab	1			*		*
8	CCE 5151	Cloud Application with JAVA and Database Lab	1			*	*	*
9	ESE 5152	Data Structures and Algorithms Lab	1			*	*	*
10	ESE 5154	Real Time Operating Systems Lab	1			*	*	*
11	MPT 5100	Mini Project - I	4	*	*	*	*	*
12	PSD 5100	Professional Skill Development - I	1	*	*			
13	AME 5202	Deep Learning	3	*		*	*	
14	BDE 5202	Modern Databases for Big Data	3	*			*	
15	ESE 5201	Device Drivers	3			*		*
16	AME 5204	Applied Computer Vision	3			*	*	
	BDE 5203	Data Engineering	3	*			*	
	CCE 5001	DevOps for Cloud	3			*	*	*



	CCE 5203	Cloud Security Essentials	3			*	*	*
	CYE 5202	Web Security and Secure Coding	3	*	*			*
	ESE 5234	Entrepreneurship	3			*	*	
17	AME 5003	Principles of Natural Language Processing	3			*	*	
	AME 5205	Advanced Generative AI and Agentic Systems	3	*		*	*	
	CCE 5241	Cloud Operations	3			*	*	*
	CYE 5203	Machine Learning for Cybersecurity	3			*	*	
	ESE 5244	IT Project Management	3			*	*	
18	AME 5252	Deep Learning Lab	1	*				*
19	BDE 5252	Modern Databases for Big Data Lab	1	*			*	*
20	ESE 5251	Device Drivers Lab	1			*		*
21	AME 5254	Applied Computer Vision Lab	1			*	*	*
	BDE 5253	Data Engineering Lab	1			*		*
	CCE 5051	DevOps for Cloud Lab	1			*	*	*
	CCE 5253	Cloud Security Essentials Lab	1			*	*	*
	CYE 5252	Web Security and Secure Coding Lab	1	*	*			*
	ESE 5284	Entrepreneurship Lab	1			*	*	
22	AME 5053	Principles of Natural Language Processing Lab	1			*		*



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

	AME 5255	Advanced Generative AI and Agentic Systems Lab	1			*	*	*
	CCE 5291	Cloud Operations Lab	1			*	*	*
	CYE 5253	Machine Learning for Cybersecurity Lab	1			*	*	*
	ESE 5294	IT Project Management Lab	1			*		*
23	MPT 5200	Mini Project - II	4	*	*	*	*	*
24	PSD 5200	Professional Skill Development - II	1			*	*	*
25	CSE 6098	Project Work	30	*	*	*	*	*